Seminar3

luni, 25 octombrie 2021 10:01

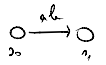
Automate finite nedeterministe

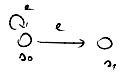
$M = (K, \Sigma, \Delta, s, F)$ — K: mulțimea de stări; Σ: alfabet; Δ: mulțime de tranziții; s: starea inițială; F: mulțimea de stări finale

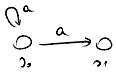
$\delta(s_0, a) = s_1$

$\Delta \subseteq (K \times \Sigma^* \times K)$ $\quad ((s_0, a, s_1) \in \Delta$

- tranziții pe șiruri
- pot să am tranziții lipsă
- tranziții pe șirul vid
- tranziții multiple

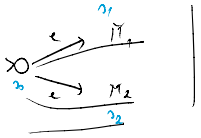


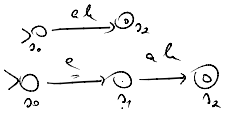




M_1 $\qquad$ M_2

01 $\qquad$ $\#_0(w) : 2$ sau $\#_1(w) \not\equiv 2$





ab

e · ab

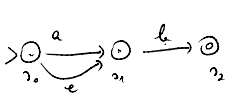
e ab

ab e

e a e b e

$\Sigma = \{a, b\}$

$L = \{ab, b\}$



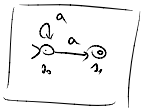
b

ab

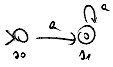
Rb.



ab · x

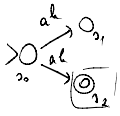


a^+



a^+

Un cuvânt este acceptat de AFN de Σ ul putem o cale cu nr egal în (s, e) unde $s \in F$.

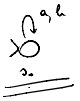


ab.

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1. $\Sigma = \{a, b\}$

$L = \emptyset$ $\qquad$ $\Sigma = \{a, b\}$



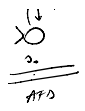
AFD



AFN

∀ AFD → AFN

∀ AFN → AFD





∀ AFD → AFN

∄ AFN → AFD



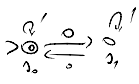
$L = \{e\}$

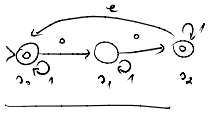
AFD : din $\forall q \in K$ am tranzitii pe $\forall x \in \Sigma$

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2. $\Sigma = \{0, 1\}$

$L = \{ w \in \Sigma^* / \#_0(w) : 2 \}$





$(00)^*$

0101

00

e

$w = 00$

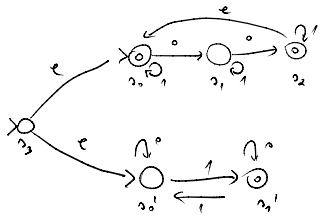
$q_0 \xrightarrow{0} q_1 \xrightarrow{0} q_2 \xrightarrow{e} q_0 \xrightarrow{0} q_1 \notin F$

$w =$ ~~000~~

$w =$ ~~00~~ 0

e

$L = \{ w \in \Sigma^* / \#_0(w) : 2$ sau $\#_1(w) \not: 2 \}$



$\#_0 : 2$

$\#_1 \not: 2.$

$w = 01$ ✓

path 1: $q_3 \xrightarrow{e} q_0 \xrightarrow{0} q_1 \xrightarrow{1} q_1 \notin F$

path 2: $q_3 \xrightarrow{e} q_0' \xrightarrow{1} q_1' \in F$

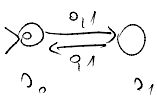


$1^* 0 1^* 0 1^*$

nu e valid

4. $L = \{ w \in \Sigma^* / |w| : 2 \}$ $\Sigma = \{0, 1\}$

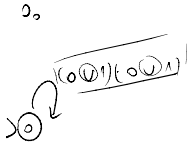
AFD / AFN



AFN



0011



$(0 \cup 1)(0 \cup 1)^*$

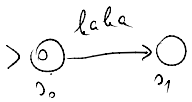
5. $\Sigma = \{a, 1\}$

$L = \{ w \in \Sigma^* / \; w \text{ e de forma } \underline{1^n}, \; n \geq 3 \}$



6. $\Sigma = \{a, b\}$

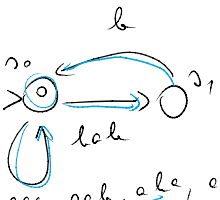
$L = \{ w \in \Sigma^* / \; \#_{\underline{baba}}(w) = 0 \}$



$\equiv$ baba $\rightarrow$ ✓ $\; 1 : q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_0 \in F$

$\rightarrow$ ✗ $\; 2 : q_0 \xrightarrow{baba} q_1 \notin F$

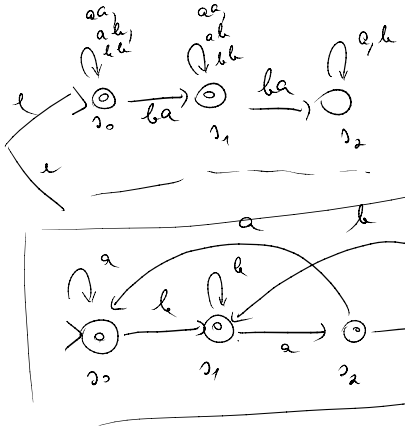
ab $\qquad (q_0, ab) \vdash^* (p, e)$



baba

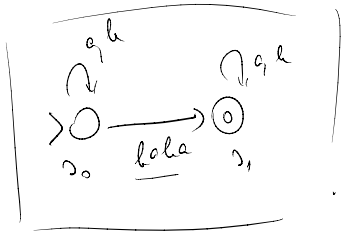
aac, aab, aba, abb, baa, bba, bbb

a $\qquad$ baba baba



aba

$L = \{ w \in \Sigma^* / \; \#_{baba}(w) > 0 \}$

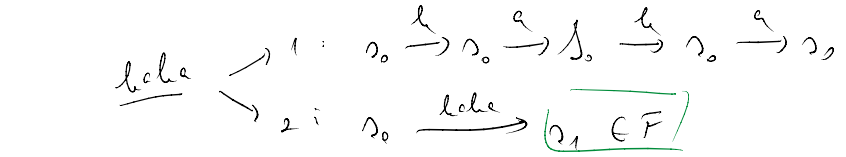


baba

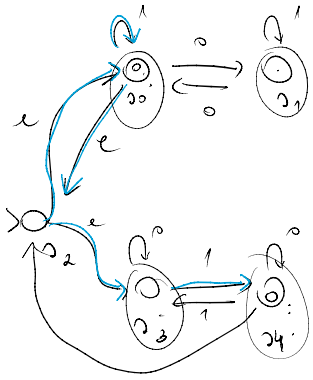


aababa!

$\rightarrow 1 : q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_0 \xrightarrow{b} q_0 \xrightarrow{a} q_0$



$L = \{ w \in \Sigma^* / \#_0(w) : 2 \quad \text{ou} \quad \#_1(w) \not: 2 \}$

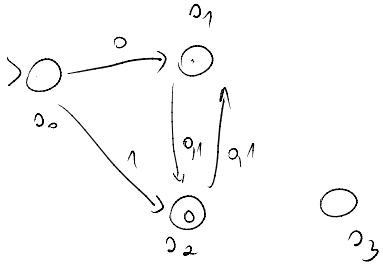


$q_2 \rightarrow$

11 !

✓



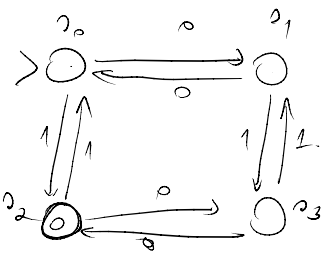


q_3

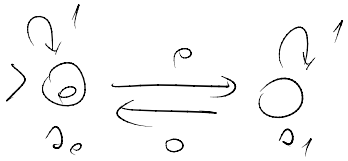
$\#_0 : 2$ $\quad$ $\#_0 \not: 2$

$\#_1 : 2$

$\#_1 \not: 2$

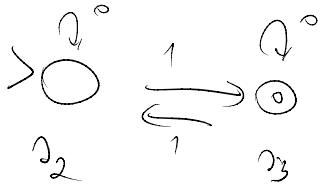


M_1



$\#_0 : 2$

M_2



$\#_1 \not:$

$K_1 = \{ v_0, v_1 \}$

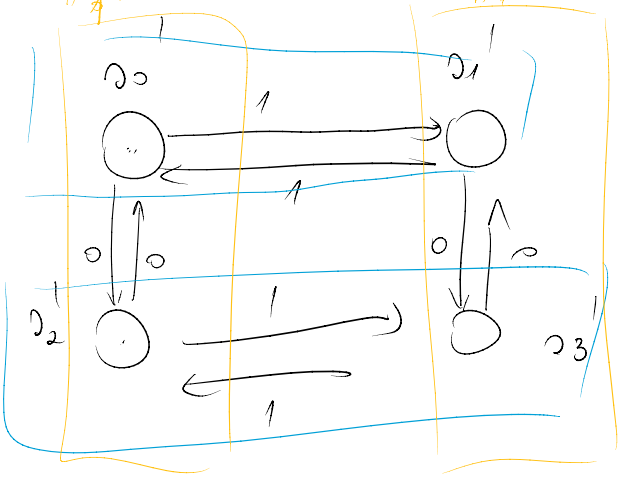
$K_2 = \{ v_2, v_3 \}$

$K_1 \times K_2$:

v_0' v_1' v_2' v_3'

$\{ (v_0, v_2), (v_0, v_3), (v_1, v_2), (v_1, v_3) \}$

$v_0' = \begin{cases} \#_0 : 2 \\ \#_1 : 2 \end{cases}$ $v_1' = \begin{cases} \#_0 : 2 \\ \#_1 \not: 2 \end{cases}$ $v_2' = \begin{cases} \#_0 \\ \#_1 \end{cases}$



$\#_0 : 2$

$\#_0 \not: 2.$

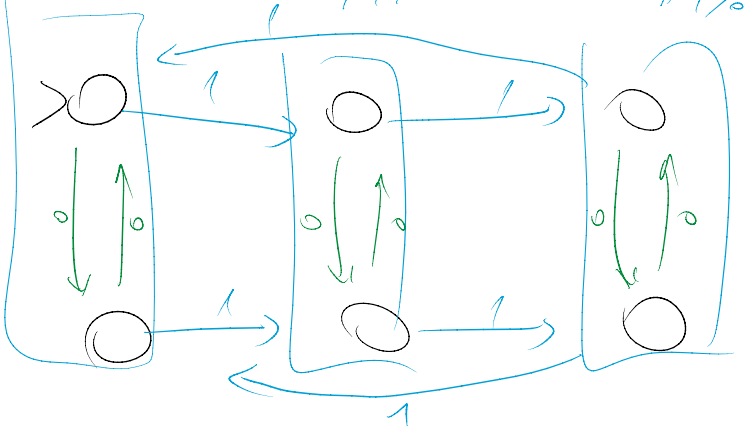
$\#_0 : 2$

2.

$\#_1 \not: 3.$

3.

$\#_1 \% 3 = 0$ $\#_1 \% 3 = 1$ $\#_1 \% 3 = 2.$



$\#_0 : 2$

$\#_0 \not: 2$